

Atomic, Molecular and Optical (AMO) Physics

Controlling, tracking and visualizing matter through the help of radiation

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High harmonic generation (HHG) and photoelectron momentum distribution (PEMD) are some of the most important processes driving the study of matter-radiation interaction and will lead to important advances in fundamental science and engineering. Through HHG, for example, it is possible to produce coherent radiations outperforming the external fields in regards to frequency and pulse duration. The current theory, valid for stationary systems, falls short at the description of nonstationary systems, since it ignores radiations induced by the external force. Following Yangaliev *et al.*[1], we aim at extending the quantum theory of radiation for real nonstationary systems, i.e in 3 dimensions and for realistic potentials.

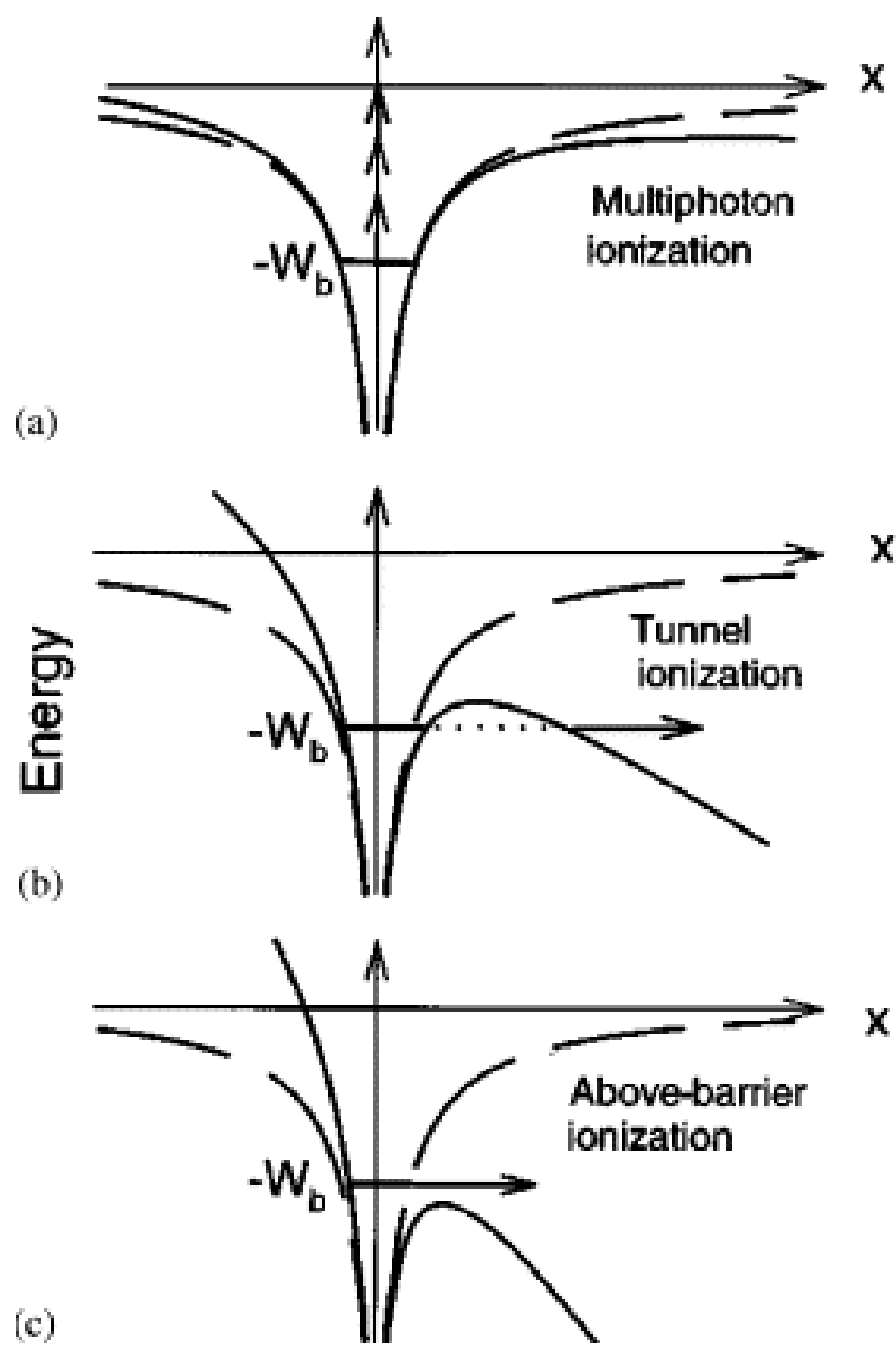
Theory

Classical physics (High school)

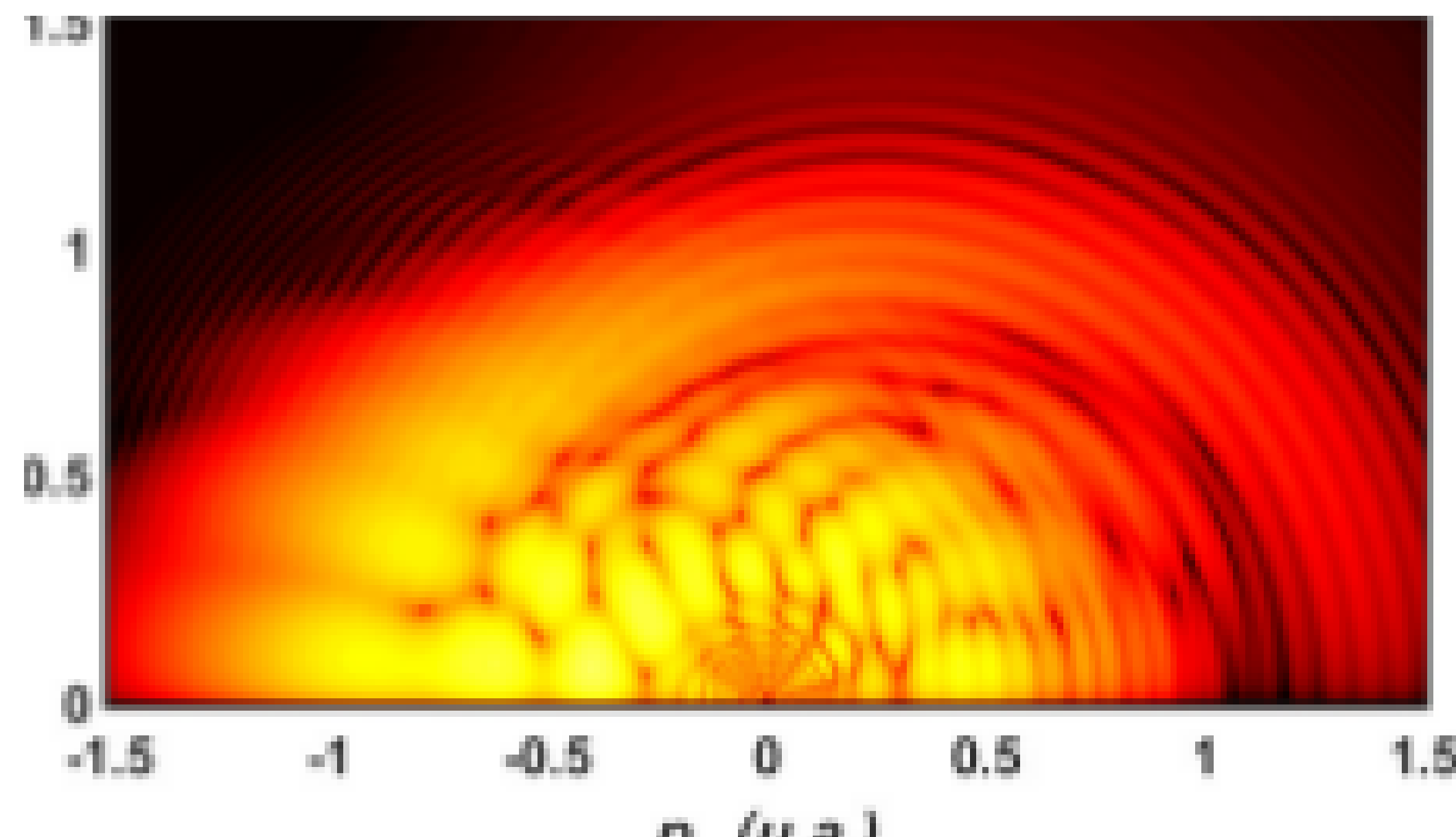
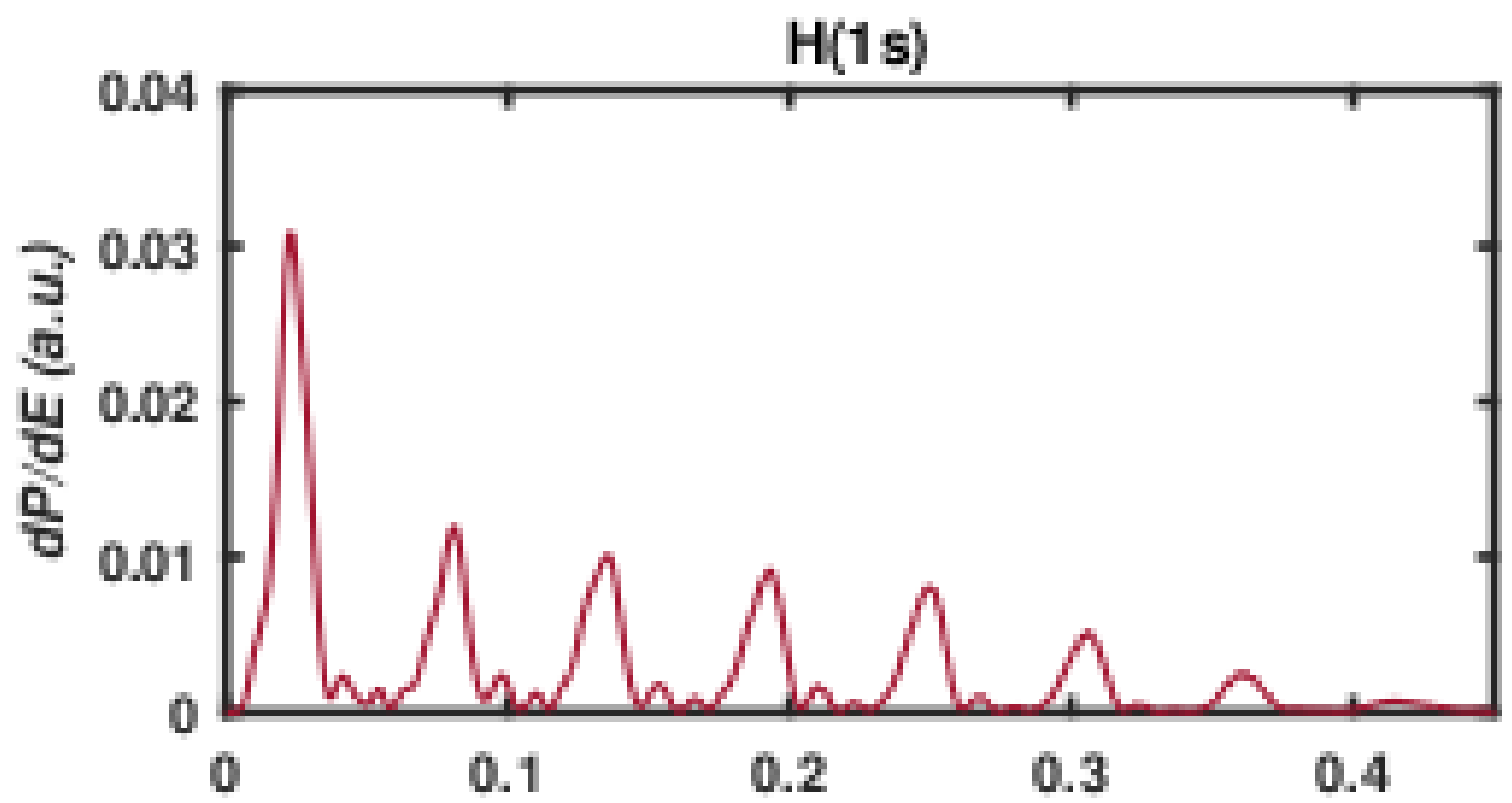
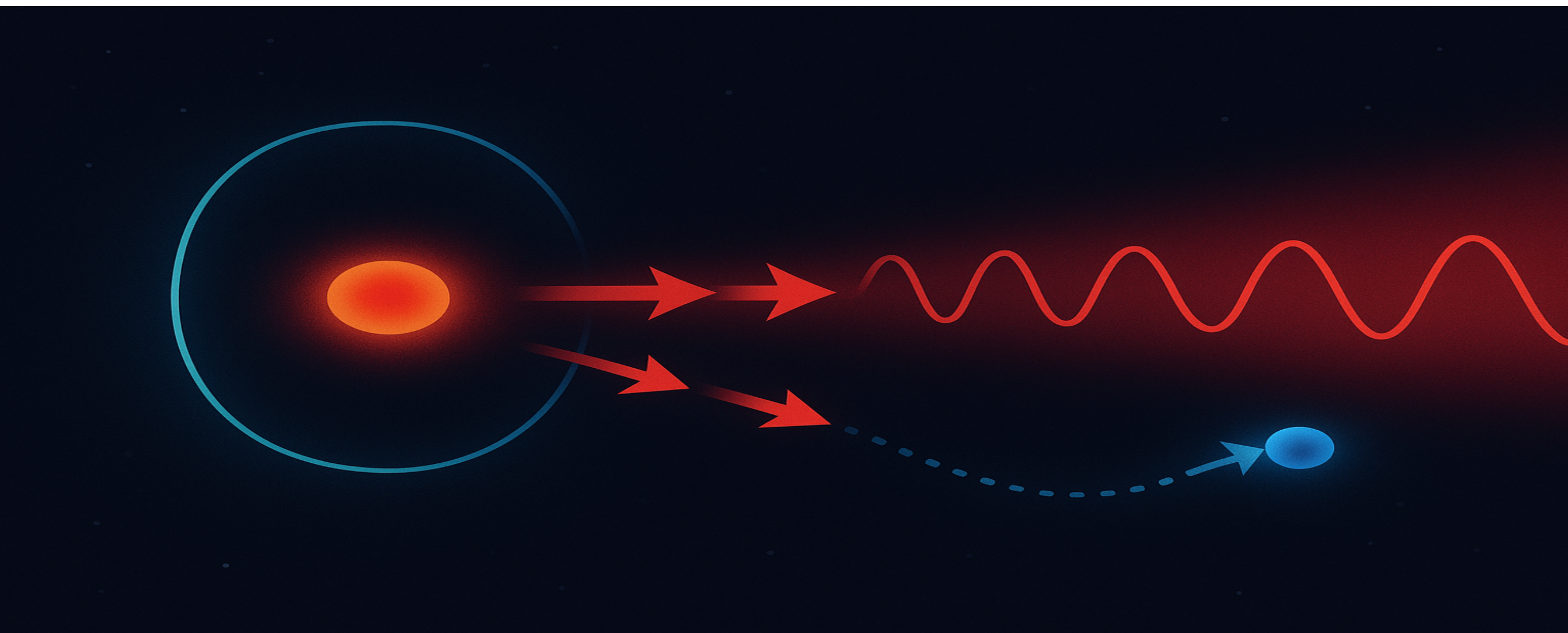
$$E = \frac{1}{2}mv^2 - \frac{e^2}{4\pi\epsilon_0 x} - e\mathbf{E} \cdot \mathbf{r} \implies i\frac{\partial}{\partial t}\psi(\mathbf{r},t) = \left(-i\hbar\frac{\partial^2}{\partial r^2} - \frac{e^2}{4\pi\epsilon_0 x} - e\mathbf{E} \cdot \mathbf{r}\right)\Psi(\mathbf{x},t)$$

Quantum physics (University)

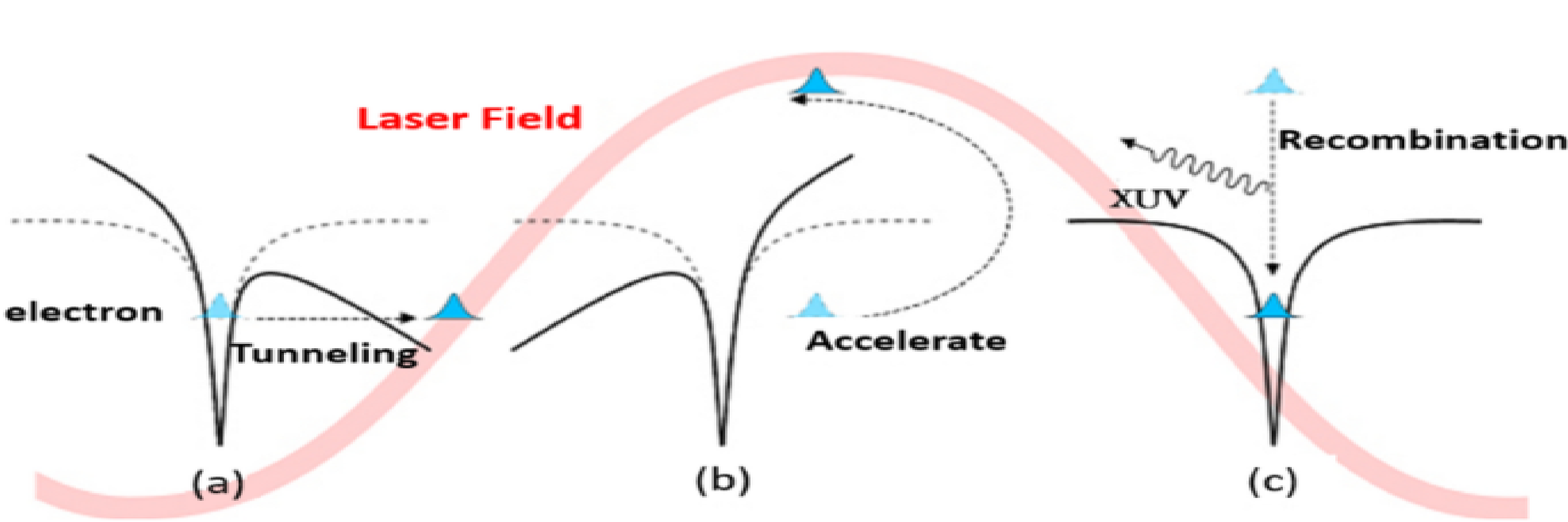
1. Ionizations



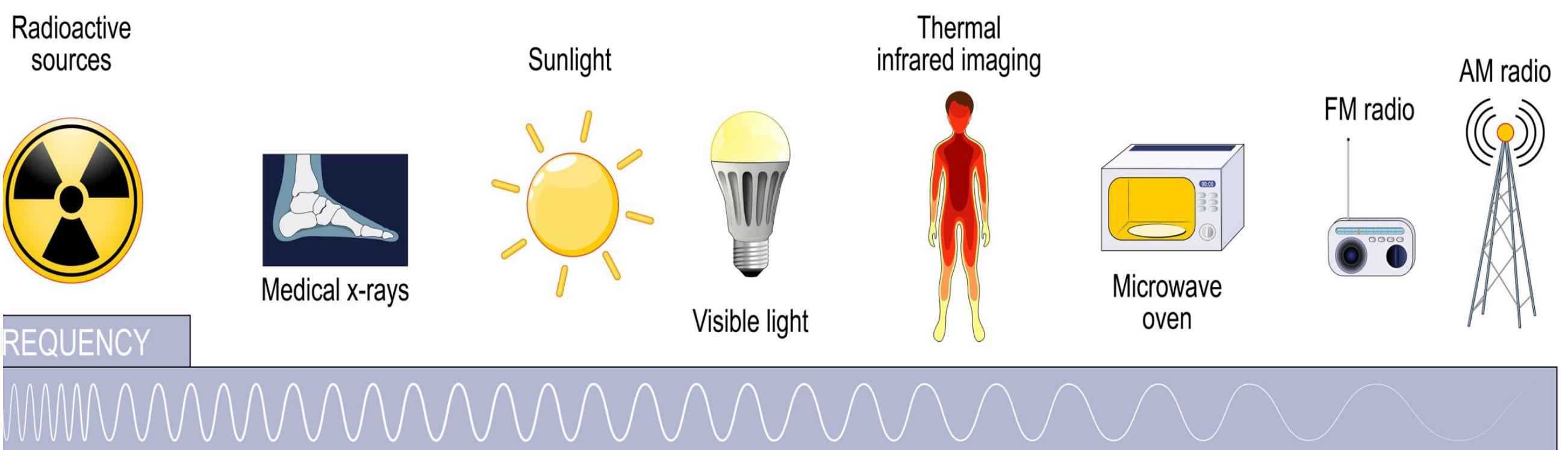
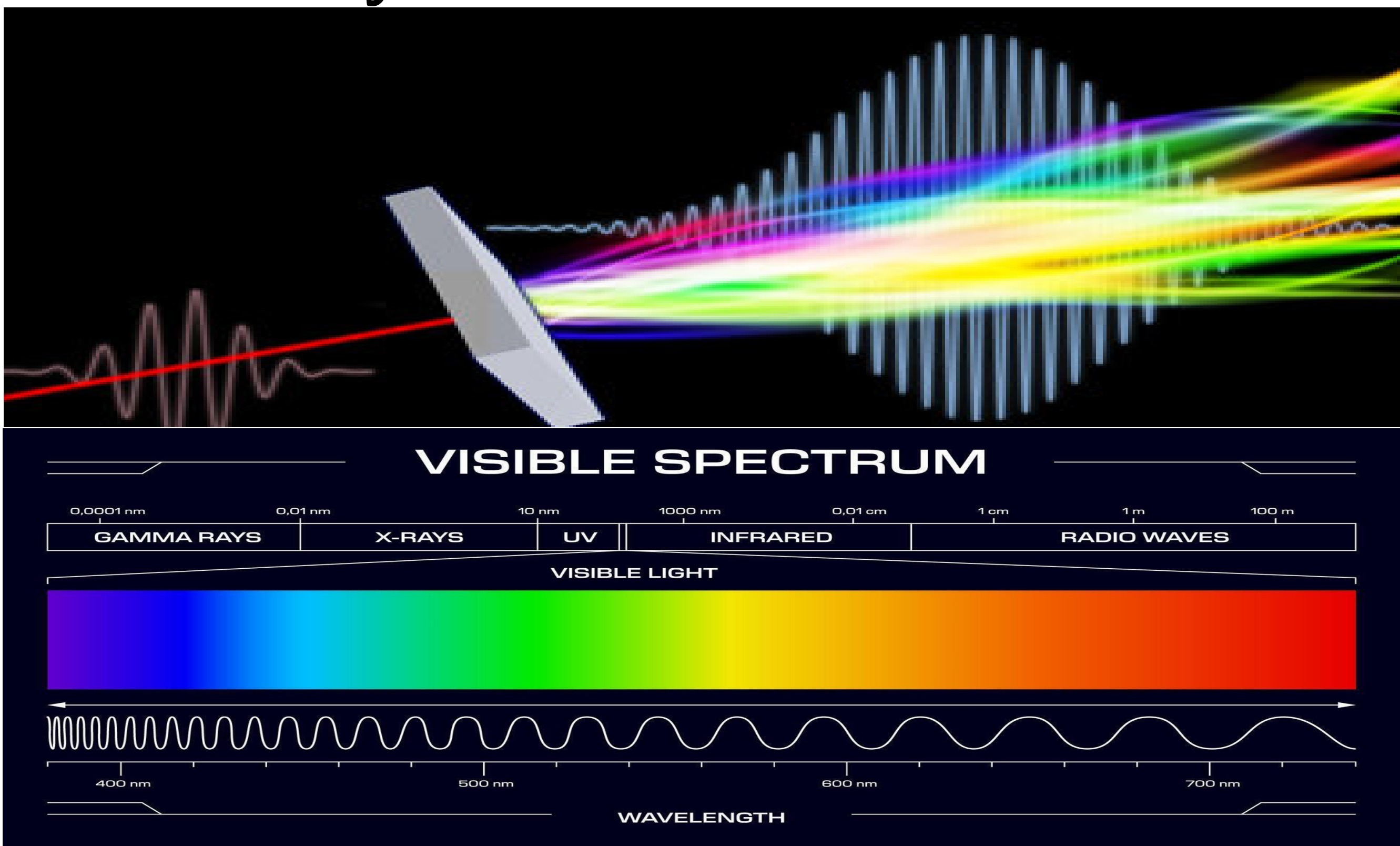
2. Multiphoton ionization



3. High Harmonic Generation



Have you ever seen a blue laser?

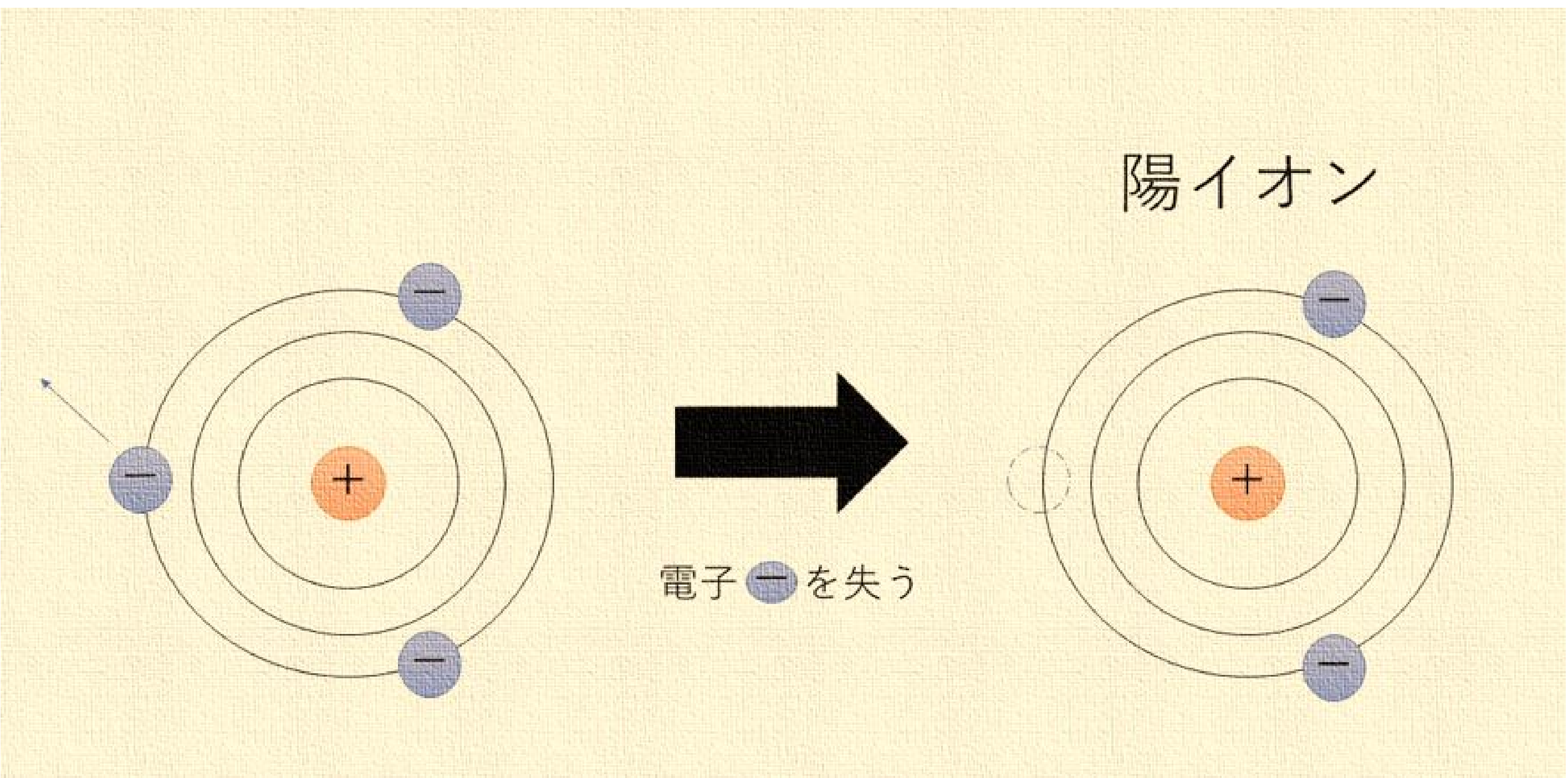


4. Interest

- From different ionizations can be obtained
 - electrons with more energy than they should (MPI)
 - a laser more energetic than the one we used (HHG)
 - electron ionized even with insufficient energy (TI)
- Multiphoton Ionization (MPI)
 - electron is ionized by many photons instead of one
 - allows to ionize even with insufficient photon energy
- High Harmonic Generation (HHG)
 - allows to multiply laser energy many fold

5. More applications

- Quantum imagery technics (MRI, LIED) allowing to scan the body or the molecules
- High resolution camera allowing to track the movement of atoms at the femtosecond scale
- Attosecond physics in general...



Temporary page!

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